

Databases (24303)

Laboratory sessions · P102 & P202

Viktoriia Popova

viktoriia.popova@upf.edu

Lab plan

Date	Lab	What we do
Thu 29 Oct	LAB01 · Session 1	Conceptual model: the Entity–Relationship diagram
Thu 5 Nov	LAB01 · Session 2	Finish the ER diagram and build the relational model
Thu 12 Nov	LAB01 · Session 3	Connect to the UPF MySQL server and create the tables
Thu 19 Nov	LAB02 · Session 1	SQL queries (SELECT)
Thu 26 Nov	LAB02 · Session 2	Functions, procedures, triggers and events
Thu 3 Dec	LAB02 · Session 3	Finish and submit LAB02
Fri 18 Dec	Final exam	11:30–14:30 · Rooms 52.019, 52.219, 52.221, 52.223

Teams of 3 students (mandatory for labs). Tools: MySQL + DBeaver, draw.io for diagrams.

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The ER alphabet



Entity · strong

A thing we store data about. Has its **own key**.



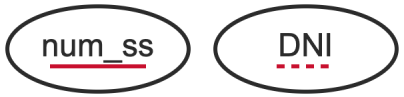
Entity · weak

Needs its **parent** to be identified: room 1 *of which* pavilion?



Attribute

One value that describes an entity.



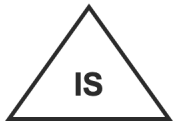
Keys

Solid = primary key
dashed = alternative key (DNI)
or partial key of a weak entity



Relationship

An **action** (a verb). **Double** = links a weak entity.



Generalization

"**is a**": a doctor *is a* person.

1 · N · M on the lines = **cardinality**: how many on each side. 1:1 · 1:N · N:M

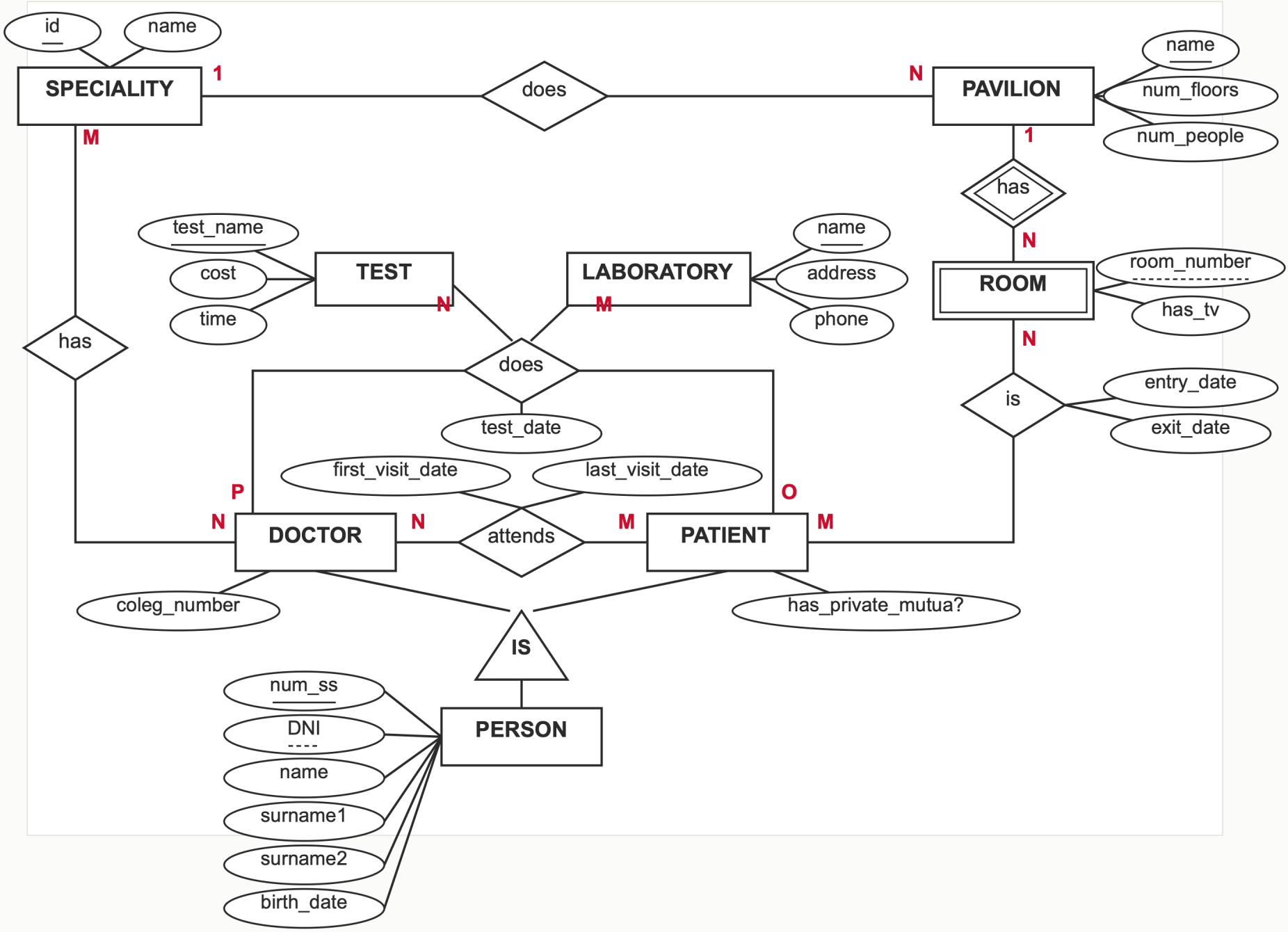
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LAB0 · Hospital Clínic

Open now on Aula Global

LABORATORY 01 → LAB0 Example

Task (PDF) · Solution (PNG) · draw.io file



From text to diagram

1 **Read**
the whole text.
No drawing.

→ **2** **Colour**
things, properties,
actions, quantities.

→ **3** **List**
a draft.
It will change.

→ **4** **Draw**
after a 3rd read.
Check every sentence.

"**One or several doctors attend** a **patient**, and we store the **date** of the first visit."

doctors, patient → entities (rectangles)

attend → relationship (diamond)

date → attribute (oval)

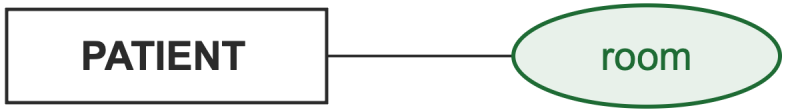
one or several → cardinality N

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Is “room” an attribute or an entity?

1st read · patient paragraph

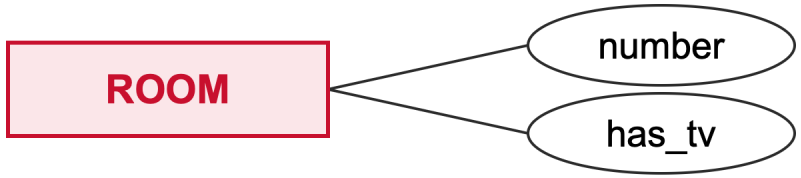
*“…in **which room** they are currently…”*



Looks like an **attribute**. OK for a first draft.

2nd read · last paragraph

*“a **set of rooms**. **For each of the rooms**: number, television”*



Many rooms + **own data** → **entity**.

ENTITY
many of them, or own data

ATTRIBUTE
just one value

NOTHING
only one in the DB (the hospital)

CLUE
“For each X…” → X is an entity

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Same data twice? Make a PERSON

1 **Find what is the same** *"For both, we will store..."*

DOCTOR

num_ss | DNI | name | surname1 | surname2 | birth_date | coleg_number

PATIENT

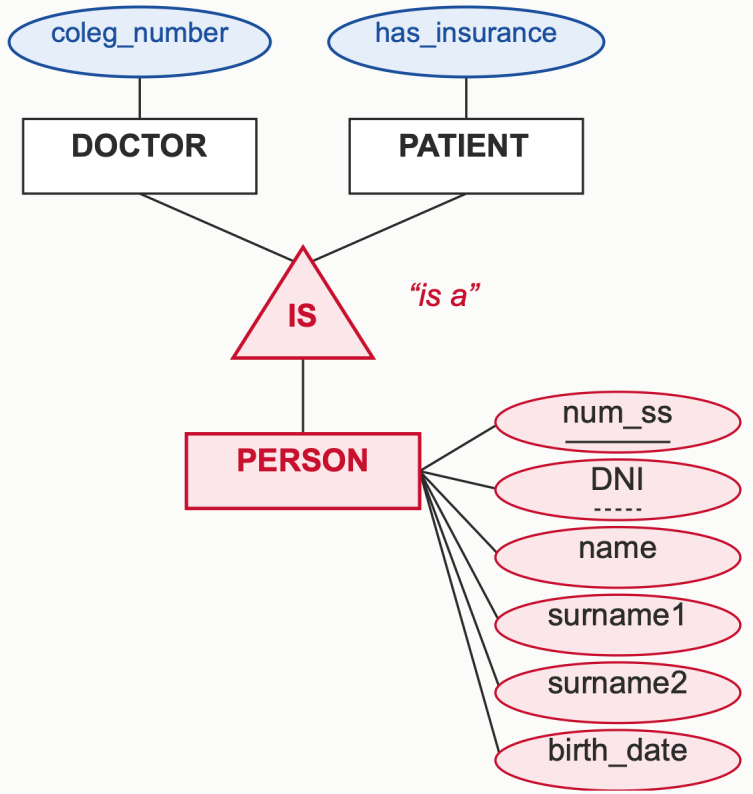
num_ss | DNI | name | surname1 | surname2 | birth_date | has_insurance

2 **Move the same data up** into **PERSON**, the **superentity**

3 **DOCTOR and PATIENT** become **subentities**: they keep only **their own** data

Triangle IS = "is a"
a doctor *is a* person

Spot it:
"for both", "all of them have"



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Two traps, and the key



Trap 1 · Don’t store what changes

María is **27** in 2026 → **wrong** in 2027.
Store **birth_date**, the age is calculated.



Trap 2 · One field = one value

“María García López” in one field:
you can’t find everyone called **García**.



The key · always unique

Never empty, never changes. Pick **one**:
num_ss (everyone has it). DNI = alternative.

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Specialities: an entity, not a text field

*"…the **specialty or specialties** they practice."*

✕ As a text attribute



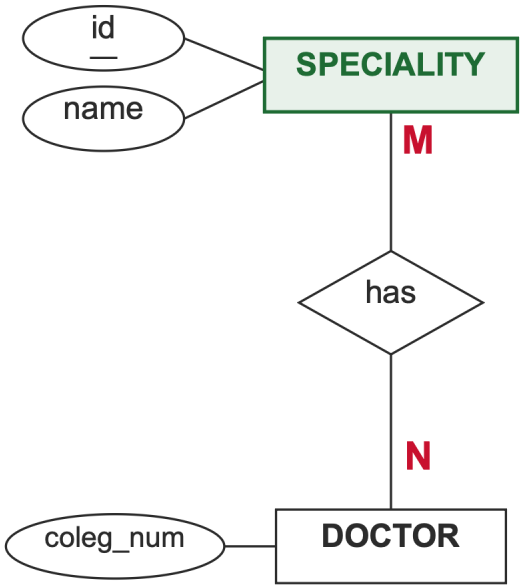
Many values in one box, spelled differently → search breaks.

Several per doctor → entity

Written **once**, same spelling for everyone

Reused later by the pavilions

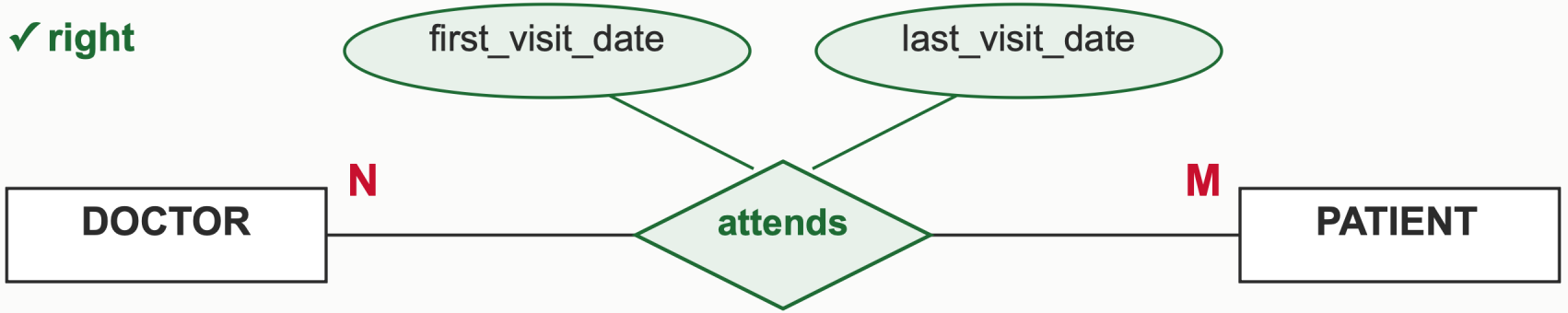
✓ As an entity, N:M

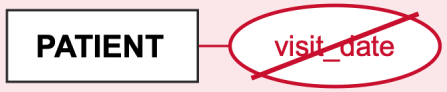


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Data about a pair goes on the relationship

✓ **right**





✕ **wrong:** Ana sees 2 doctors. Whose date is it?

Same trap: "from the patient… date of admission and discharge" → on **is** (ROOM–PATIENT), as **entry_date**, **exit_date**

"One year" → first + last visit is enough

Ask: does it change if I change the doctor **or** the patient? → on the **diamond**

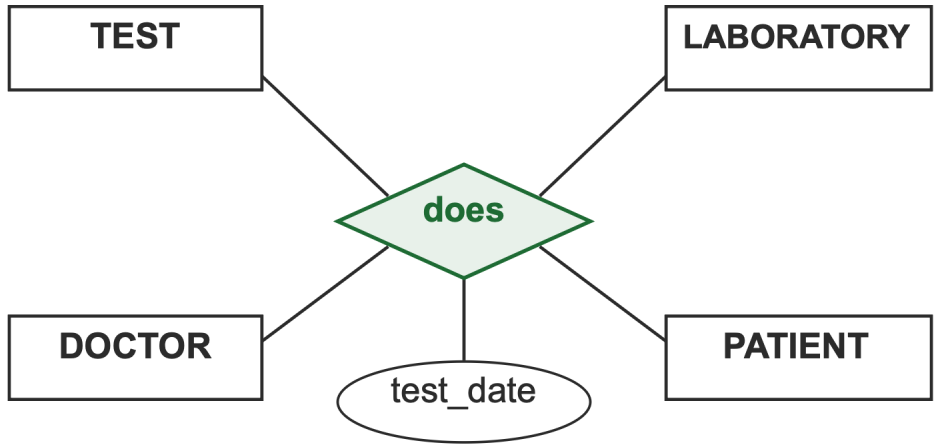
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Tests and pavilions

A · One test = 4 entities

✕ **3 separate lines:** Ana's test, Lab A or Lab B?

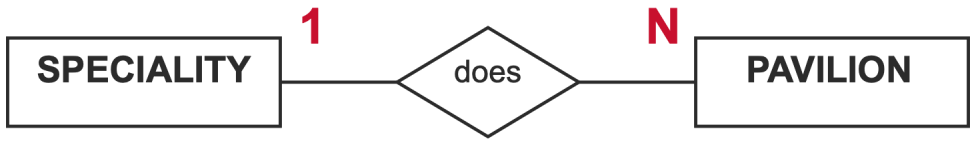
✓ **1 diamond:** who · for whom · what · where



B · Read the numbers

"Each pavilion: **a single** specialty" → **1**

"**Several** pavilions, same specialty" → **N**



Trick: say it both ways.
1 specialty → **N** pavilions
1 pavilion → **1** specialty

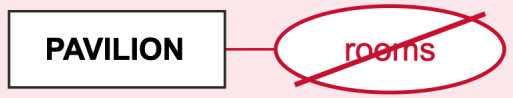
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ROOM: a weak entity

*"The room number will **start at 1 for each pavilion**."*

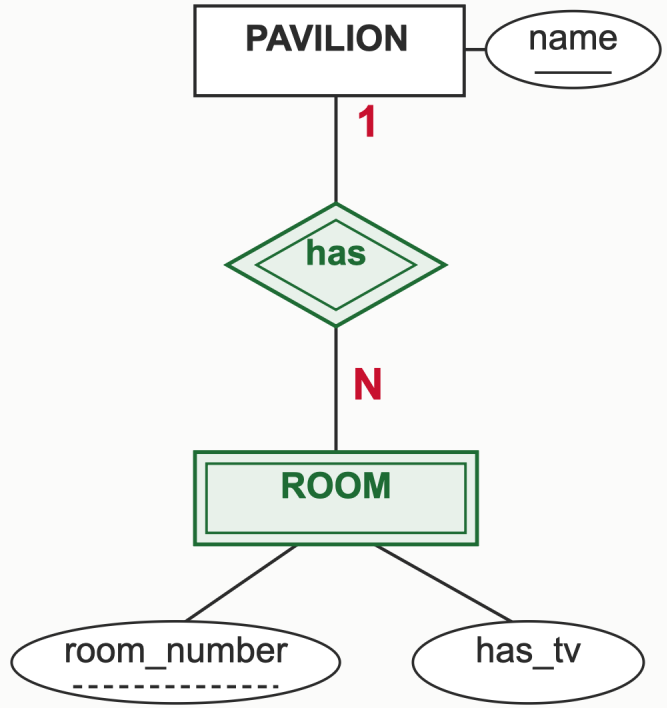
Cardio · **room 1** | Surgery · **room 1** **same number!**

"Room 1" alone is not enough → **pavilion + room_number**



✕ attribute: where do the TV and the patients go?

double rectangle	**double** diamond	**dashed** key



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An entity related to itself

Coming up in LAB01

Imagine: *"some doctors **supervise other doctors**"*

✕ A new entity BOSS



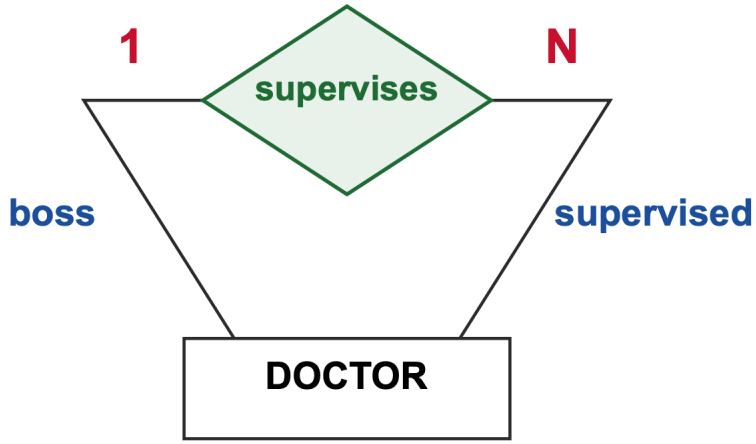
The boss **is also a doctor** → same data twice.

Clue: "some X … **other X**"

Draw: one rectangle, **two lines** to the diamond

Write the roles on the lines: who is who?

✓ Recursive relationship

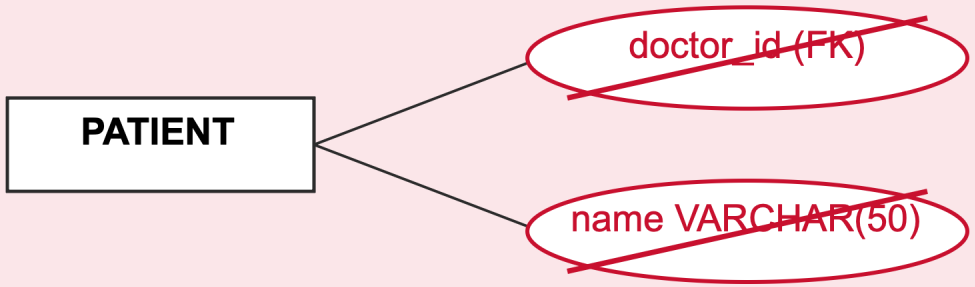


1 boss supervises N doctors

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Don’t mix ER and tables

✕ Table stuff in the ER



Foreign keys and data types are **not** in the conceptual model.

✓ The diamond is the link



No ids of other entities. The FK appears **later**, from the diamond.

Class 1–2 · Conceptual (ER)
entities · attributes · relationships · cardinality · keys

Class 2–3 · Relational (tables)
tables · PK · **FK** · data types · SQL

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Checklist for LAB01

1. **Many, or own data?**
   → entity
2. **About a pair?**
   → on the relationship
3. **Can be calculated?**
   → don't store it
4. **Several values in one field?**
   → split it
5. **Number repeats per parent?**
   → weak entity
6. **Same attributes twice?**
   → generalization (IS)
7. **Related to itself?**
   → recursive relationship
8. **History needed?**
   → usually N:M

⚠ Common mistakes

- Foreign keys in the ER diagram (later!)
- Messy delivery: keep it clean and readable
- Age or full name as attributes (calculated / non-atomic)
- Forgot weak entities, IS or recursive links

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TODAY

LAB01 · Session 1

1. Sit with your team of 3
2. Read the LAB01 task on Aula Global
3. Start the ER diagram in draw.io
4. Ask questions — I'll come round to each team